Area, Volume, 3D Solids & Modeling Strategy

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • [FRQ] • Target: Rectangular Prism Volume, Surface Area & Space Diagonal

A rectangular prism has length $l = 8\,\text{cm}$, width $w = 6\,\text{cm}$, and height $h = 5\,\text{cm}$, as shown. Find (a) its volume, (b) its total surface area, and (c) the exact length of its space diagonal d.

(a) $V = lwh = 8(6)(5) = 240\,\text{cm}^3$. (b) $S = 2(lw + lh + wh) = 2(48 + 40 + 30) = 236\,\text{cm}^2$. (c) $d = \sqrt{8^2 + 6^2 + 5^2} = \sqrt{64 + 36 + 25} = \sqrt{125} = 5\sqrt{5}\,\text{cm}$.

2 [Foundational] • [FRQ] • Target: Cylinder Volume & Surface Area

A right circular cylinder has base radius $r = 4\,\text{cm}$ and height $h = 10\,\text{cm}$. Compute its exact volume, lateral surface area, and total surface area in terms of π.

Volume $V = \pi r^2 h = \pi(16)(10) = 160\pi\,\text{cm}^3$. Lateral surface area $L = 2\pi rh = 80\pi\,\text{cm}^2$. Total surface area $S = 80\pi + 2\pi(16) = 112\pi\,\text{cm}^2$.

3 [Foundational] • [FRQ] • Target: Right Circular Cone Geometry

A right circular cone has base radius $r = 5\,\text{m}$ and vertical height $h = 12\,\text{m}$. Find (a) its slant height ℓ, (b) its exact volume, and (c) its total surface area in terms of π.

(a) Slant height $\ell = \sqrt{5^2 + 12^2} = 13\,\text{m}$. (b) Volume $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(25)(12) = 100\pi\,\text{m}^3$. (c) Total surface area $S = \pi r^2 + \pi r\ell = 25\pi + 65\pi = 90\pi\,\text{m}^2$.

4 [Foundational] • [FRQ] • Target: Sphere Surface Area & Volume

A sphere has surface area $144\pi\,\text{cm}^2$. Find its radius r and its exact volume V in terms of π.

Since $S = 4\pi r^2 = 144\pi$, we have $r^2 = 36 \Rightarrow r = 6\,\text{cm}$. Therefore the volume is $V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(216) = 288\pi\,\text{cm}^3$.

5 [Foundational] • [FRQ] • Target: Square Pyramid Volume & Slant Height

A right square pyramid has base edge length $b = 16\,\text{cm}$ and vertical height $h = 15\,\text{cm}$. Find (a) its volume, (b) the slant height ℓ of each triangular face, and (c) its total surface area.

(a) $V = \frac{1}{3}b^2 h = \frac{1}{3}(256)(15) = 1{,}280\,\text{cm}^3$. (b) Half the base edge is $8\,\text{cm}$, so slant height $\ell = \sqrt{8^2 + 15^2} = 17\,\text{cm}$. (c) Total surface area $= 16^2 + 4 \cdot \frac{1}{2}(16)(17) = 256 + 544 = 800\,\text{cm}^2$.

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6 [Foundational] • [FRQ] • Target: Composite L-Shaped Polygons

Compute the area and perimeter of the L-shaped polygon with vertices $(0,0), (10,0), (10,4), (4,4), (4,9), (0,9)$.

Split into a 4×9 vertical rectangle (area 36) and a 6×4 horizontal extension (area 24), giving Total Area $= 36 + 24 = 60$. The perimeter equals that of its 10×9 bounding box: $2(10 + 9) = 38$.

7 [Foundational] • [FRQ] • Target: Rhombus Diagonals, Area & Perimeter

A rhombus has diagonals of lengths $d_1 = 16\,\text{cm}$ and $d_2 = 30\,\text{cm}$. Find the area and perimeter of the rhombus.

Area $= \frac{1}{2}d_1 d_2 = \frac{1}{2}(16)(30) = 240\,\text{cm}^2$. Since the diagonals bisect each other perpendicularly into legs of $8\,\text{cm}$ and $15\,\text{cm}$, each side has length $\sqrt{8^2 + 15^2} = 17\,\text{cm}$, and the perimeter is $4(17) = 68\,\text{cm}$.

Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

8 [Intermediate] • [FRQ] • Target: Volume Conservation Across Recast Solids

A solid metal sphere of radius $6\,\text{cm}$ is melted down and recast into a solid right circular cone of base radius $12\,\text{cm}$. Find the height h of the cone.

By conservation of volume, $V_{\text{sphere}} = V_{\text{cone}}$: $\frac{4}{3}\pi(6)^3 = \frac{1}{3}\pi(12)^2 h \Rightarrow 4(216) = 144h \Rightarrow 864 = 144h \Rightarrow h = 6\,\text{cm}$.

9 [Intermediate] • [FRQ] • Target: Annular / Hollow Cylinder Volume

A cylindrical pipe of length $20\,\text{cm}$ has outer radius $R = 5\,\text{cm}$ and inner hollow radius $r = 3\,\text{cm}$. Find the exact volume of metal in the pipe wall.

$V = \pi(R^2 - r^2)h = \pi(5^2 - 3^2)(20) = \pi(16)(20) = 320\pi\,\text{cm}^3$.

10 [Intermediate] • [FRQ] • Target: Composite 3D Solids (Cylinder + Hemisphere)

A grain silo consists of a right circular cylinder of radius $3\,\text{m}$ and height $10\,\text{m}$ topped by a hemispherical dome of radius $3\,\text{m}$. Find the exact total volume of the silo in terms of π.

Cylinder volume $= \pi(3^2)(10) = 90\pi\,\text{m}^3$. Hemisphere volume $= \frac{2}{3}\pi(3^3) = 18\pi\,\text{m}^3$. Total volume $= 90\pi + 18\pi = 108\pi\,\text{m}^3$.

11 [Intermediate] • [FRQ] • Target: Inscribed Sphere in a Cube

A sphere is inscribed inside a cube of edge length $8\,\text{cm}$ so that the sphere is tangent to all 6 faces. Find the exact volume of the empty space inside the cube but outside the sphere.

Cube volume $= 8^3 = 512\,\text{cm}^3$. The inscribed sphere has diameter $8\,\text{cm}$, so $r = 4\,\text{cm}$ and $V_{\text{sphere}} = \frac{4}{3}\pi(4^3) = \frac{256\pi}{3}\,\text{cm}^3$. Empty volume $= 512 - \frac{256\pi}{3}\,\text{cm}^3$.

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12 [Intermediate] • [FRQ] • Target: Net of a Cone from a Circular Sector

A circular sector of radius $10\,\text{cm}$ and arc length $12\pi\,\text{cm}$ is rolled up to form the lateral surface of a right circular cone (without overlap). Find the base radius r and vertical height h of the resulting cone.

The arc length of the sector becomes the base circumference of the cone: $2\pi r = 12\pi \Rightarrow r = 6\,\text{cm}$. The sector radius $10\,\text{cm}$ becomes the slant height $\ell = 10\,\text{cm}$, so the vertical height is $h = \sqrt{10^2 - 6^2} = 8\,\text{cm}$.

13 [Intermediate] • [FRQ] • Target: Hexagonal Prism Volume & Surface Area

A regular hexagonal prism has base side length $4\,\text{cm}$ and height $10\,\text{cm}$. Find its exact volume and total surface area.

Base area of a regular hexagon of side $s = 4$ is $B = 6 \cdot \frac{\sqrt{3}}{4}(4^2) = 24\sqrt{3}\ \text{cm}^2$. Volume $= Bh = 240\sqrt{3}\ \text{cm}^3$. Total surface area $= 2B + 6(4)(10) = 48\sqrt{3} + 240\,\text{cm}^2$.

14 [Intermediate] • [FRQ] • Target: Solids of Revolution of Polygons

A right triangle with legs $6\,\text{cm}$ and $8\,\text{cm}$ is revolved 360° about its $8\,\text{cm}$ leg. Describe the resulting 3D solid and compute its exact volume and lateral surface area.

Revolving the right triangle about its $8\,\text{cm}$ leg generates a right circular cone of height $h = 8\,\text{cm}$, base radius $r = 6\,\text{cm}$, and slant height $\ell = 10\,\text{cm}$. Volume $= \frac{1}{3}\pi(6^2)(8) = 96\pi\,\text{cm}^3$, and lateral surface area $= \pi r\ell = 60\pi\,\text{cm}^2$.

Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • [FRQ] • Target: 3D Box Construction via Quadratic Equations • Cross-Lesson: Topic 3.01 (Quadratics)

An open-top rectangular box is constructed by cutting congruent squares of side $2\,\text{cm}$ from the four corners of a rectangular sheet of tin whose length is $5\,\text{cm}$ greater than its width, and folding up the flaps. If the volume of the box is $100\,\text{cm}^3$, find the original dimensions of the tin sheet.

Let the original width be w and length be $w + 5$. After cutting $2\,\text{cm}$ squares from each corner, the base dimensions are $(w - 4)$ and $(w + 1)$ with height $2\,\text{cm}$. Volume:

$2(w-4)(w+1) = 100 \Rightarrow w^2 - 3w - 4 = 50 \Rightarrow w^2 - 3w - 54 = 0 \Rightarrow (w-9)(w+6) = 0$. Thus $w = 9\,\text{cm}$ and length $= 14\,\text{cm}$.

16 [Challenging] • [FRQ] • Target: Similar Cones & Cubic Volume Ratio • Cross-Lesson: Topic 7.01 (Similar Triangles) & Topic 5.01 (Ratios)

A conical water tank (vertex pointing down) has top radius $R = 6\,\text{m}$ and total height $H = 12\,\text{m}$. When the water in the tank has depth $h = 8\,\text{m}$, find the radius r of the water surface and the fraction of the tank's total volume that is filled.

By similar triangles, $\frac{r}{h} = \frac{R}{H} = \frac{6}{12} = \frac{1}{2}$, so at $h = 8\,\text{m}$ the surface radius is $r = 4\,\text{m}$. Since the water cone is similar to the full tank with linear ratio $\frac{h}{H} = \frac{8}{12} = \frac{2}{3}$, the filled volume fraction is $\left(\frac{2}{3}\right)^3 = \frac{8}{27}$.

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17 [Challenging] • [FRQ] • Target: Cube Inscribed in a Sphere • Cross-Lesson: Topic 7.01 (3D Pythagorean Theorem)

A cube is inscribed inside a sphere of radius $R = 3\sqrt{3}$ cm so that all 8 vertices of the cube lie on the surface of the sphere. Find the edge length s and volume V of the cube.

The space diagonal of the cube equals the diameter of the sphere: $s\sqrt{3} = 2R = 6\sqrt{3} \Rightarrow s = 6\,\text{cm}$. Thus the volume of the cube is $V = 6^3 = 216\,\text{cm}^3$.

18 [Challenging] • [FRQ] • Target: Cylinder Inscribed in a Sphere • Cross-Lesson: Topic 9.01 (Circles & Chords)

A right circular cylinder is inscribed in a sphere of radius 5 cm. If the height of the cylinder is 6 cm, find the base radius r and exact volume of the cylinder.

A cross-section through the axis of the cylinder and center of the sphere shows a right triangle with hypotenuse $R = 5$ cm and legs $h/2 = 3$ cm and r. By the Pythagorean theorem, $r = \sqrt{5^2 - 3^2} = 4$ cm. Therefore $V = \pi r^2 h = \pi(16)(6) = 96\pi\,\text{cm}^3$.

19 [Challenging] • [FRQ] • Target: Volumetric Flow Rate & Dimensional Modeling • Cross-Lesson: Topic 5.01 (Rates & Proportions)

Water flows through a cylindrical pipe of internal diameter 4 cm at a linear speed of 50 cm/s into an empty rectangular tank of base 100 cm by 60 cm. How many seconds (in terms of π) does it take to fill the tank to a depth of 20 cm?

Volumetric flow rate $Q = \pi r^2 v = \pi(2^2)(50) = 200\pi\,\text{cm}^3/\text{s}$. Required tank volume $= 100 \times 60 \times 20 = 120{,}000\,\text{cm}^3$. Time $t = \frac{120{,}000}{200\pi} = \frac{600}{\pi}$ seconds.

20 [Challenging] • [FRQ] • Target: 3D Regular Tetrahedron Altitude & Volume • Cross-Lesson: Topic 7.01 (Equilateral Triangles & Centroids)

Find the exact surface area and volume of a regular tetrahedron (all 4 faces are equilateral triangles) of edge length $s = 6$ cm.

Each equilateral face has area $\frac{\sqrt{3}}{4}(6^2) = 9\sqrt{3}$ cm^2, so total surface area is $4(9\sqrt{3}) = 36\sqrt{3}$ cm^2. The distance from a base vertex to the base centroid is $\frac{2}{3}(3\sqrt{3}) = 2\sqrt{3}$, so the altitude is $h = \sqrt{6^2 - (2\sqrt{3})^2} = \sqrt{24} = 2\sqrt{6}$ cm, giving volume $V = \frac{1}{3}(9\sqrt{3})(2\sqrt{6}) = 18\sqrt{2}$ cm^3.

Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

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21 [Foundational] • [MCQ] • Target: Cube Surface Area to Volume

What is the volume of a cube whose total surface area is $150\,\text{cm}^2$?

a. $25\,\text{cm}^3$

b. $100\,\text{cm}^3$

c. $125\,\text{cm}^3$ ✓

d. $216\,\text{cm}^3$

Since $6s^2 = 150$, we have $s^2 = 25 \Rightarrow s = 5\,\text{cm}$. Thus $V = s^3 = 125\,\text{cm}^3$.

22 [Foundational] • [MCQ] • Target: Cylinder Volume & Base Circumference

A right circular cylinder has volume 72π and height 8. What is the circumference of its circular base?

a. 3π

b. 6π ✓

c. 9π

d. 12π

From $\pi r^2(8) = 72\pi$, we get $r^2 = 9 \Rightarrow r = 3$. The base circumference is $2\pi r = 6\pi$.

23 [Foundational] • [MCQ] • Target: Area of a Trapezoid

What is the area of a trapezoid with parallel bases of lengths 9 and 15 and height 7?

a. 42

b. 84 ✓

c. 105

d. 168

Area $= \frac{b_1 + b_2}{2} \cdot h = \frac{9 + 15}{2} \cdot 7 = 12 \cdot 7 = 84$.

24 [Foundational] • [MCQ] • Target: Comparing Cone & Cylinder Volumes

A right circular cone and a right circular cylinder have the same base radius and the same volume. The height of the cone is how many times the height of the cylinder?

a. $\frac{1}{3}$

b. 1

c. 3 ✓

d. 9

Since $\frac{1}{3}\pi r^2 h_{\text{cone}} = \pi r^2 h_{\text{cyl}}$, we have $h_{\text{cone}} = 3h_{\text{cyl}}$.

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25 [Foundational] • [MCQ] • Target: Cubic Scaling of Sphere Volume

If the radius of a sphere is doubled, its volume is multiplied by what factor?

a. 2

b. 4

c. 6

d. 8 ✓

Because $V = \frac{4}{3}\pi r^3$, replacing r with $2r$ multiplies the volume by $2^3 = 8$.

26 [Foundational] • [MCQ] • Target: Rectangle Perimeter & Area

A rectangular field has perimeter $100\,\text{m}$ and length $32\,\text{m}$. What is the area of the field?

a. $544\,\text{m}^2$

b. $576\,\text{m}^2$ ✓

c. $640\,\text{m}^2$

d. $1024\,\text{m}^2$

Since $2(l + w) = 100 \Rightarrow 32 + w = 50 \Rightarrow w = 18\,\text{m}$. Area $= 32 \times 18 = 576\,\text{m}^2$.

27 [Foundational] • [MCQ] • Target: Volume of a Triangular Prism

What is the volume of a right triangular prism whose base is a right triangle with legs 5 and 12 and whose prism height is 10?

a. 150

b. 300 ✓

c. 600

d. 650

Base area $B = \frac{1}{2}(5)(12) = 30$. Volume $= Bh = 30 \times 10 = 300$.

28 [Foundational] • [MCQ] • Target: Volume of a Sphere

A solid sphere of radius $r = 3$ is shown below. What is its exact volume?

a. 12π

b. 27π

c. 36π ✓

d. 108π

$V = \frac{4}{3}\pi(3^3) = 36\pi$.

Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

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29 [Intermediate] • [MCQ] • Target: Additive Volumes of Spheres ($3^3 + 4^3 + 5^3 = 6^3$)

Three solid lead spheres of radii $3\,\text{cm}$, $4\,\text{cm}$, and $5\,\text{cm}$ are melted together and cast into a single large sphere. What is the radius of the new sphere?

a. $6\,\text{cm}$ ✓ **b.** $7\,\text{cm}$

c. $8\,\text{cm}$ **d.** $12\,\text{cm}$

By volume conservation, $R^3 = 3^3 + 4^3 + 5^3 = 27 + 64 + 125 = 216 \Rightarrow R = \sqrt[3]{216} = 6\,\text{cm}$.

30 [Intermediate] • [MCQ] • Target: 3D Space Diagonal

What is the length of the longest straight rod that can fit completely inside a rectangular box measuring $3\,\text{ft}$ by $4\,\text{ft}$ by $12\,\text{ft}$?

a. $12\,\text{ft}$ **b.** $13\,\text{ft}$ ✓

c. $15\,\text{ft}$ **d.** $19\,\text{ft}$

The longest rod lies along the 3D space diagonal: $d = \sqrt{3^2 + 4^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13\,\text{ft}$.

31 [Intermediate] • [MCQ] • Target: Archimedes Displacement in a Cylinder

A cylindrical glass of radius $3\,\text{cm}$ contains water to a depth of $8\,\text{cm}$. When a solid glass marble of radius $3\,\text{cm}$ is completely submerged in the water, by how many centimeters does the water level rise?

a. $2\,\text{cm}$ **b.** $3\,\text{cm}$

c. $4\,\text{cm}$ ✓ **d.** $6\,\text{cm}$

The submerged sphere displaces volume $V = \frac{4}{3}\pi(3^3) = 36\pi\,\text{cm}^3$. In a cylinder of base area $\pi(3^2) = 9\pi\,\text{cm}^2$, the water height rises by $\Delta h = \frac{36\pi}{9\pi} = 4\,\text{cm}$.

32 [Intermediate] • [MCQ] • Target: Working Backward from Pyramid Volume

A square-based pyramid has volume $400\,\text{cm}^3$ and height $12\,\text{cm}$. What is the perimeter of its square base?

a. $20\,\text{cm}$ **b.** $40\,\text{cm}$ ✓

c. $80\,\text{cm}$ **d.** $100\,\text{cm}$

From $\frac{1}{3}b^2(12) = 400$, we get $4b^2 = 400 \Rightarrow b^2 = 100 \Rightarrow b = 10\,\text{cm}$. The perimeter of the square base is $4b = 40\,\text{cm}$.

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33 [Intermediate] • [MCQ] • Target: Cylinder Formed from a Rectangular Sheet

A rectangular sheet of paper of dimensions 6π cm by 10 cm is rolled into a cylinder of height 10 cm so that the 6π cm edges meet without overlap. What is the volume of this cylinder?

a. $30\pi\,\text{cm}^3$

b. $60\pi\,\text{cm}^3$

c. $90\pi\,\text{cm}^3$ ✓

d. $360\pi\,\text{cm}^3$

The circumference is $2\pi r = 6\pi \Rightarrow r = 3$ cm. Thus the volume is $\pi r^2 h = \pi(9)(10) = 90\pi\,\text{cm}^3$.

34 [Intermediate] • [MCQ] • Target: Surface-Area-to-Volume Ratio of Similar Solids

The ratio of the surface areas of two cubes is $16:49$. What is the ratio of their volumes?

a. $4:7$

b. $16:49$

c. $64:343$ ✓

d. $256:2401$

Edge lengths have ratio $\sqrt{16}:\sqrt{49} = 4:7$, so volumes have ratio $4^3:7^3 = 64:343$.

35 [Intermediate] • [MCQ] • Target: Parallelogram Area via Trigonometry

A parallelogram has adjacent sides of lengths 10 and 14, and the acute angle between them is 30°. What is the area of the parallelogram?

a. 35

b. 70 ✓

c. $70\sqrt{3}$

d. 140

The altitude to the base of length 14 is $h = 10\sin 30^\circ = 5$. Thus Area $= 14 \times 5 = 70$.

36 [Intermediate] • [MCQ] • Target: Volume of a Conical Frustum

A frustum is formed by slicing off the top third (by height) of a right circular cone of volume $135\,\text{cm}^3$ with a plane parallel to its base. What is the volume of the remaining bottom frustum?

a. $90\,\text{cm}^3$

b. $115\,\text{cm}^3$

c. $120\,\text{cm}^3$

d. $130\,\text{cm}^3$ ✓

The removed top cone has height $\frac{1}{3}H$, so its volume is $\left(\frac{1}{3}\right)^3(135) = \frac{135}{27} = 5\,\text{cm}^3$. The remaining frustum has volume $135 - 5 = 130\,\text{cm}^3$.

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37 [Intermediate] • [MCQ] • Target: Total Surface Area of a Solid Hemisphere

A solid hemisphere of radius R has total surface area (including its flat circular base) equal to 75π. What is R?

a. 3

b. 5 ✓

c. $5\sqrt{3}$

d. 15

A solid hemisphere has curved area $2\pi R^2$ plus flat circular base πR^2, giving $3\pi R^2 = 75\pi \Rightarrow R^2 = 25 \Rightarrow R = 5$.

Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Isoperimetric Optimization via Quadratics • Cross-Lesson: Topic 3.01 (Quadratic Optimization)

Among all rectangles with a fixed perimeter of $48\,\text{cm}$, what is the maximum possible area?

a. $120\,\text{cm}^2$

b. $135\,\text{cm}^2$

c. $144\,\text{cm}^2$ ✓

d. $256\,\text{cm}^2$

With $2(l + w) = 48 \Rightarrow l + w = 24$, the area $A(w) = w(24 - w) = -(w - 12)^2 + 144$ is maximized when the rectangle is a 12×12 square of area $144\,\text{cm}^2$.

39 [Challenging] • [MCQ] • Target: Archimedes' Sphere-to-Cylinder Ratio • Cross-Lesson: Topic 5.01 (Ratios)

A sphere is inscribed in a right circular cylinder so that the sphere touches the top base, bottom base, and lateral wall. What is the ratio of the volume of the sphere to the volume of the cylinder?

a. $1:2$

b. $2:3$ ✓

c. $3:4$

d. $4:3$

The cylinder has radius r and height $h = 2r$, so $V_{\text{cyl}} = \pi r^2(2r) = 2\pi r^3$. Since $V_{\text{sphere}} = \frac{4}{3}\pi r^3$, the ratio is $\frac{4/3}{2} = \frac{2}{3}$ (Archimedes' theorem).

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40 [Challenging] • [MCQ] • Target: Volume from Pairwise Face Areas • Cross-Lesson: Topic 4.01 (Radicals & Algebraic Products)

In a rectangular prism, the areas of three mutually adjacent faces are $12\,\text{cm}^2$, $15\,\text{cm}^2$, and $20\,\text{cm}^2$. What is the volume of the prism?

a. $47\,\text{cm}^3$

b. $60\,\text{cm}^3$ ✓

c. $120\,\text{cm}^3$

d. $180\,\text{cm}^3$

Given $lw = 12$, $wh = 15$, and $lh = 20$, multiply all three equations:

$(lwh)^2 = 12 \times 15 \times 20 = 3600 \Rightarrow V = lwh = \sqrt{3600} = 60\,\text{cm}^3$.

41 [Challenging] • [MCQ] • Target: Coordinate Solids of Revolution • Cross-Lesson: Topic 1.01 (Coordinate Geometry)

A region in the first quadrant bounded by $x = 0$, $y = 0$, $x = 4$, and $y = 3$ is revolved 360° about the y-axis. What is the volume of the resulting solid?

a. 36π

b. 48π ✓

c. 64π

d. 144π

Revolving the 4×3 rectangle about the y-axis generates a cylinder of radius $r = 4$ and height $h = 3$, whose volume is $\pi(4^2)(3) = 48\pi$.

42 [Challenging] • [MCQ] • Target: Algebraic Surface-to-Volume Equating • Cross-Lesson: Topic 4.01 (Polynomial Equations)

The surface area of a cube is numerically equal to twice its volume. What is the length of the space diagonal of the cube?

a. 3

b. $3\sqrt{2}$

c. $3\sqrt{3}$ ✓

d. $6\sqrt{3}$

Set $6s^2 = 2s^3 \Rightarrow s = 3$. The space diagonal of a cube of edge $s = 3$ is $s\sqrt{3} = 3\sqrt{3}$.

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43 [Challenging] • [MCQ] • Target: Tilted Cylinder Volume Symmetry • Cross-Lesson: Topic 7.01 (45-45-90 Triangles)

A right circular cylinder of radius $6\,\text{cm}$ and height $15\,\text{cm}$ is initially completely full of water, and is then tilted at a 45° angle until water stops spilling out. What volume of water was spilled?

a. $108\pi\,\text{cm}^3$

b. $180\pi\,\text{cm}^3$

c. $216\pi\,\text{cm}^3$ ✓

d. $324\pi\,\text{cm}^3$

At a 45° tilt, the empty air wedge at the top of the cylinder spans across the diameter $2r = 12\,\text{cm}$ and therefore has vertical height $12\tan 45^\circ = 12\,\text{cm}$. By symmetry, the air wedge occupies exactly half of the top $12\,\text{cm}$ cylindrical section: $V_{\text{spilled}} = \frac{1}{2}\pi(6^2)(12) = 216\pi\,\text{cm}^3$.

44 [Challenging] • [MCQ] • Target: Composite Area of Polygons & Circular Sectors • Cross-Lesson: Topic 9.01 (Circular Sectors)

An equilateral triangle of side length 6 has circular sectors of radius 3 centered at each of its three vertices removed. What is the area of the remaining central region?

a. $9\sqrt{3} - \frac{9\pi}{2}$ ✓

b. $9\sqrt{3} - 9\pi$

c. $18\sqrt{3} - \frac{9\pi}{2}$

d. $36\sqrt{3} - 9\pi$

The equilateral triangle has area $\frac{\sqrt{3}}{4}(6^2) = 9\sqrt{3}$. Each vertex has a 60° sector of radius 3, so the three sectors together form a 180° semicircle of area $\frac{1}{2}\pi(3^2) = \frac{9\pi}{2}$. Thus the remaining area is $9\sqrt{3} - \frac{9\pi}{2}$.

45 [Challenging] • [MCQ] • Target: Percentage Scaling of Surface Area • Cross-Lesson: Topic 5.01 (Percentages)

If the edge length of a cube is increased by 10%, by what percentage does its total surface area increase?

a. 10%

b. 20%

c. 21% ✓

d. 33.1%

Surface area scales as the square of the linear scale factor: $(1.10)^2 = 1.21$, which corresponds to a 21% increase.

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